

BLOCK ISLAND STATE, RI, USA

WMO: 725058

Lat: 41.168N

Lon: 71.578W

Elev: 32

StdP: 100.94

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94793

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.3	-21.3	0.6	-9.4	-18.2	0.8	-7.0	14.1	3.2	12.6	2.6	6.5	330	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	27.8	22.4	27.0	22.0	25.9	21.3	24.0	26.5	23.3	25.5	22.5	24.5	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	25.1	22.5	17.3	24.6	22.1	16.9	24.3	72.0	26.7	69.3	25.7	66.5	24.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.5	9.2	-14.1	30.3	2.3	1.9	-15.8	31.7	-17.1	32.8	-18.4	33.8	-20.1	35.2		
(5)			-15.2	24.7	2.4	1.1	-16.9	25.5	-18.3	26.1	-19.6	26.7	-21.3	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	0.4	1.0	3.6	8.2	13.0	18.2	22.1	21.8	18.8	13.5	8.3	4.0		
	DBStd	8.40	5.26	4.41	3.76	2.96	3.00	2.95	2.16	2.27	2.67	3.51	4.10	4.21		
	HDD10.0	1107	300	253	201	67	6	0	0	0	0	10	79	192		
	HDD18.3	2941	557	487	457	303	168	37	1	2	27	153	302	446		
	CDD10.0	1515	1	0	2	14	98	247	376	364	263	118	27	4		
	CDD18.3	306	0	0	0	0	2	34	119	108	39	3	0	0		
	CDH23.3	970	0	0	0	1	7	89	446	370	57	1	0	0		
	CDH26.7	118	0	0	0	0	0	10	64	42	3	0	0	0		
(14) Wind	WSAvg	5.1	5.8	5.6	5.6	5.3	4.9	4.7	4.5	4.2	4.4	5.1	5.5	5.6		
(15) Precipitation	PrecAvg	1024	83	81	95	91	84	70	68	81	88	74	105	100		
	PrecMax	1514	224	175	204	234	155	220	180	228	292	186	231	206		
	PrecMin	612	6	20	27	21	36	14	8	4	8	21	23	24		
	PrecStd	194	50	41	45	50	40	53	42	53	64	35	58	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	11.4	14.8	18.4	23.9	27.8	30.9	29.9	27.0	22.6	17.8	15.0		
		MCWB	11.8	10.0	10.4	13.1	17.5	21.5	24.0	23.7	21.8	19.0	15.8	13.4		
	2%	DB	11.4	9.6	12.1	15.9	21.4	26.0	28.0	27.7	24.9	21.1	16.9	13.0		
		MCWB	10.5	8.0	8.8	10.8	16.5	20.2	22.5	22.6	21.0	18.5	14.8	11.6		
	5%	DB	9.0	8.0	10.1	13.7	18.9	23.8	27.2	26.6	23.6	19.6	15.9	12.3		
		MCWB	8.0	6.4	7.8	9.8	15.1	19.2	22.2	21.9	20.5	17.8	13.8	11.1		
	10%	DB	7.6	7.2	8.7	12.5	17.5	22.5	26.0	25.7	22.6	18.3	14.1	11.0		
		MCWB	6.0	5.1	6.8	9.1	14.0	18.6	21.5	21.5	20.0	16.2	12.1	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	10.7	11.6	14.3	19.1	22.8	24.9	25.1	23.6	20.3	16.9	14.0		
		MCDB	12.8	11.0	13.0	17.0	22.1	25.8	28.5	27.9	25.8	21.4	17.5	14.8		
	2%	WB	10.6	8.4	9.7	12.5	17.4	21.4	24.0	24.1	22.5	19.2	15.3	12.4		
		MCDB	11.3	9.0	11.4	14.2	19.9	24.3	26.8	26.4	23.9	20.3	16.5	13.1		
	5%	WB	8.1	7.2	8.1	11.2	16.0	20.4	23.4	23.4	21.6	18.0	14.1	11.1		
		MCDB	8.7	8.0	9.1	12.8	18.4	22.8	25.8	25.4	22.9	19.0	15.4	12.2		
	10%	WB	6.4	5.2	6.9	9.8	14.7	19.4	22.6	22.6	20.7	16.9	12.9	9.2		
		MCDB	7.5	6.8	8.3	11.9	17.0	21.6	24.8	24.5	22.1	18.1	14.3	10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	6.0	6.2	6.3	6.3	6.3	6.0	5.6	5.7	5.7	5.7		
		MCDBR	7.6	7.0	7.5	7.8	8.8	8.3	7.6	7.0	6.4	6.2	6.0	6.7		
	5% WB	MCWBR	8.0	6.7	5.9	5.1	5.2	4.7	4.0	3.8	4.3	5.1	6.1	7.4		
		MCWBR	7.5	7.3	7.1	7.0	7.5	7.1	6.4	5.8	5.5	5.6	5.7	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.321	0.343	0.377	0.424	0.463	0.488	0.449	0.381	0.352	0.321	0.310		
	taud		2.497	2.473	2.448	2.388	2.290	2.223	2.160	2.266	2.428	2.509	2.550	2.549		
	Ebn at Noon		844	892	908	895	856	820	794	816	856	846	830	821		
	Edh at Noon		75	88	101	115	130	140	148	129	103	85	71	66		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.74	3.82	4.94	5.48	6.00	6.11	5.50	4.48	3.04	2.05	1.55		
	RadStd		0.15	0.28	0.29	0.53	0.53	0.59	0.33	0.32	0.34	0.28	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+53

Nomenclature: See separate page